

like documents, dictionaries, objects, and graphs. It stores the relationships in the form of a graph database.

The following commands can be helpful for using OrientDB on your Ubuntu machine:

```
$sudo apt-get update
$wget -O orientdb-community-2.2.20.tar.gz http://orientdb.com/download.php?file=orientdb-community-2.2.20.tar.gz&os=linux
$tar -zxvf orientdb-community-2.2.20.tar.gz
$sudo mv ~/orientdb-community-2.2.20 /opt/orientdb
```

CouchDB

CouchDB is an important NoSQL based open source database server developed in Erlang. It uses many protocols and formats to transfer, process and share data. JSON files are used to store the data, MapReduce is used as the base and JavaScript is used as the querying language.

Advantages

- It can store any kind of data.
- MapReduce helps in increasing its efficiency.
- The overall structure is very simple.
- Indexing and retrieval is fast.

The following commands can help you to use CouchDB on your Ubuntu machine:

```
$echo "deb https://apache.bintray.com/couchdb-deb focal main" >> /etc/apt/sources.list
$sudo apt-get update
$sudo apt install apache2 couchdb -y
```

Firebird

Firebird is an open source database management system that mainly works on relational databases. It is compatible with all operating systems like Linux, Windows and MacOS. It was originally forked from the Interbase repository, which was also an open source database.

Advantages

- The functionality of the database is not limited.
- It is a very stable and powerful database.
- The configuration and usage of the database is much simpler than other databases.

The following commands can help you use Firebird on your Ubuntu machine:

```
$sudo apt-get update
$sudo apt-get install firebird2.5-superclassic
```

Cassandra

Cassandra is owned and developed by Apache. This highly scalable, distributed, high performance database can handle large amounts of data and works really well. As it is distributed among many servers, it has no single point of failure. It is basically a NoSQL database, which implies it is not a relational database.

Advantages

- High performance
- High scalability
- Peer-to-peer architecture is used instead of master slave architecture

The following commands can be helpful in using Firebird on your Ubuntu machine:

```
$curl https://www.apache.org/dist/cassandra/KEYS | sudo apt-key add -
$sudo apt-get update
$sudo apt-get install cassandra
$sudo systemctl enable cassandra
$sudo systemctl start cassandra
```

PostgreSQL

Today PostgreSQL is one of the most popular open source relational database management systems. It is easily extensible and, at the same time, is compliant with SQL. More than 30 years of active development has gone into this DBMS.

Advantages

- In Postgres we can store a wider

variety of data compared to MySQL.

- It is largely compliant with the SQL standard.
 - It is also highly expandable.
- The following commands can be helpful for using PostgreSQL on your Ubuntu machine:

```
$sudo apt-get update
$sudo apt install postgresql postgresql-contrib
```

CockroachDB

CockroachDB is a database built for reliability, i.e., it can survive any kind of adverse situation (just like cockroaches can survive any disastrous situation and multiply). This database can handle large amounts of data. Multiclust databases can also be built.

Advantages

- Deployment is easy
- High consistency
- Transactions are distributed
- Availability is high
- It is also compatible with SQL

Redis

Redis is an open source, key value based data storage database. This NoSQL database is really handy because it uses different keys of various types.

We've gone through the most well-known and popular open source database management systems that are being used to store and manage data. Learning about these different databases can be a lot of fun. Try the different options, and use the one that fits your requirements the best. Also, do check out the official documentation of these databases. 

 By: Jishnu Saurav Mittapalli

The author is a full stack Web developer, interested in modern technologies like artificial intelligence and machine learning. He is currently working in the field of natural language processing.